Wind-Tunnel Airspeed Uncertainty

From Measurement Physics to Instrument Selection

Student Theory and Design Guide — ASEN 3501 Tier 2 Aerodynamics Lab

Central engineering question. If an experiment requires a particular airspeed and a specified measurement uncertainty, can the existing wind-tunnel instrumentation meet that requirement? If not, what would need to change, and what capability might be lost?

1 What you should be able to do

After working through this guide, you should be able to:

1. Derive airspeed from differential pressure, temperature, and atmospheric pressure.
2. Calculate the *sensitivity coefficients* used in uncertainty propagation, showing each algebraic step.
3. Explain why pressure-related and density-related contributions to *absolute* airspeed uncertainty change differently with speed.
4. Distinguish absolute standard uncertainty, u_V (m/s), from normalized standard uncertainty, u_V/V (dimensionless).
5. Convert an experimental uncertainty requirement into a pressure-sensor requirement and evaluate the resulting operating range.

2 Airspeed is a calculated measurement

For low-speed flow under the assumptions of steady, approximately incompressible flow and an appropriate total/static pressure measurement, Bernoulli's relation gives

$$P_0 = P_s + \tfrac{1}{2}\rho V^2, \tag{1}$$

$$\Delta P \equiv P_0 - P_s = \tfrac{1}{2}\rho V^2. \tag{2}$$

Solving for airspeed,

$$\boxed{V = \sqrt{\frac{2\Delta P}{\rho}}}. \tag{3}$$

For air approximated as an ideal gas,

$$\rho = \frac{P_{\text{atm}}}{RT}. \tag{4}$$

Substituting this expression for ρ into Eq. (3),

$$V = \sqrt{\frac{2\Delta P}{P_{\text{atm}}/(RT)}} \tag{5}$$

$$= \boxed{\sqrt{\frac{2\Delta P R T}{P_{\text{atm}}}}}. \tag{6}$$

Here ΔP is differential (dynamic) pressure in Pa, P_{atm} is *absolute* atmospheric pressure in Pa, T is absolute temperature in K, and $R \approx 287\,\text{J/(kg K)}$ is the specific gas constant for dry air. The airspeed V is in m/s.

Measurement-model assumptions. We treat R as known, and initially treat the three measured inputs as independent. This is a first-order uncertainty model; it does not account for all possible tunnel-flow effects, probe alignment errors, or instrument biases. We will also assume atmospheric conditions are approximately constant while varying tunnel speed.

3 How input uncertainty becomes airspeed uncertainty

Let $u_{\Delta P}$, u_T, and $u_{P_{\text{atm}}}$ denote the *standard uncertainties* of the three measured inputs (not necessarily manufacturers' quoted "accuracy" values). Under the independent-input, first-order approximation,

$$\boxed{u_V^2 = \left(\frac{\partial V}{\partial \Delta P} u_{\Delta P}\right)^2 + \left(\frac{\partial V}{\partial T} u_T\right)^2 + \left(\frac{\partial V}{\partial P_{\text{atm}}} u_{P_{\text{atm}}}\right)^2}. \tag{7}$$

Each partial derivative is a *sensitivity coefficient*: it tells us how strongly the calculated airspeed responds to a small change in that input, while the other inputs are held fixed.

3.1 Sensitivity to differential pressure ΔP

Write Eq. (6) using exponents, collecting the quantities held constant when differentiating with respect to ΔP:

$$V = \left(\frac{2RT}{P_{\text{atm}}}\right)^{1/2} (\Delta P)^{1/2}. \tag{8}$$

Apply the power rule:

$$\frac{\partial V}{\partial \Delta P} = \left(\frac{2RT}{P_{\text{atm}}}\right)^{1/2} \left[\frac{1}{2}(\Delta P)^{-1/2}\right] \tag{9}$$

$$= \frac{1}{2}\left(\frac{2RT}{P_{\text{atm}}}\right)^{1/2} \frac{(\Delta P)^{1/2}}{\Delta P} \qquad \left[(\Delta P)^{-1/2} = \frac{(\Delta P)^{1/2}}{\Delta P}\right] \tag{10}$$

$$= \frac{1}{2\Delta P} \underbrace{\left(\frac{2RT}{P_{\text{atm}}}\right)^{1/2} (\Delta P)^{1/2}}_{V}. \tag{11}$$

Thus,

$$\boxed{\frac{\partial V}{\partial \Delta P} = \frac{V}{2\Delta P}}. \tag{12}$$

Why does this sensitivity get smaller at higher speed? Equation (12) has V in its numerator, but ΔP itself grows with V^2. Substitute $\Delta P = \frac{1}{2}\rho V^2$:

$$\frac{\partial V}{\partial \Delta P} = \frac{V}{2(\frac{1}{2}\rho V^2)} \tag{13}$$

$$= \frac{V}{\rho V^2} \tag{14}$$

$$= \boxed{\frac{1}{\rho V}}. \tag{15}$$

Therefore, the *differential-pressure contribution* to absolute airspeed uncertainty is

$$\boxed{u_{V,\Delta P} = \left|\frac{\partial V}{\partial \Delta P}\right| u_{\Delta P} = \frac{u_{\Delta P}}{\rho V}}. \tag{16}$$

If $u_{\Delta P}$ and ρ are approximately constant as tunnel speed changes, then $u_{V,\Delta P} \propto 1/V$: the pressure contribution *decreases* as airspeed increases. This is a trend evaluated across operating points; it does not mean ΔP is independent of V in the experiment.

3.2 Sensitivity to temperature T

Write Eq. (6) as

$$V = (2R\Delta P)^{1/2} T^{1/2} P_{\text{atm}}^{-1/2}. \tag{17}$$

Hold R, ΔP, and P_{atm} fixed, and apply the power rule:

$$\frac{\partial V}{\partial T} = (2R\Delta P)^{1/2} \left(\frac{1}{2} T^{-1/2}\right) P_{\text{atm}}^{-1/2} \tag{18}$$

$$= \frac{1}{2}(2R\Delta P)^{1/2} \left(\frac{T^{1/2}}{T}\right) P_{\text{atm}}^{-1/2} \qquad \left[T^{-1/2} = \frac{T^{1/2}}{T}\right] \tag{19}$$

$$= \frac{1}{2T} \underbrace{(2R\Delta P)^{1/2} T^{1/2} P_{\text{atm}}^{-1/2}}_{V}. \tag{20}$$

Thus,

$$\boxed{\frac{\partial V}{\partial T} = \frac{V}{2T}}, \qquad \boxed{u_{V,T} = \frac{V}{2}\frac{u_T}{T}}. \tag{21}$$

If u_T/T is approximately constant, $u_{V,T} \propto V$: a fixed *percentage* uncertainty corresponds to a larger *absolute* uncertainty in m/s when speed is larger.

3.3 Sensitivity to atmospheric pressure P_{atm}

Again,

$$V = (2R\Delta P)^{1/2} T^{1/2} P_{\text{atm}}^{-1/2}. \tag{22}$$

Hold R, ΔP, and T fixed. Apply the power rule:

$$\frac{\partial V}{\partial P_{\text{atm}}} = (2R\Delta P)^{1/2} T^{1/2} \left(-\frac{1}{2} P_{\text{atm}}^{-3/2}\right) \tag{23}$$

$$= -\frac{1}{2}(2R\Delta P)^{1/2} T^{1/2} \left(\frac{P_{\text{atm}}^{-1/2}}{P_{\text{atm}}}\right) \qquad \left[P_{\text{atm}}^{-3/2} = \frac{P_{\text{atm}}^{-1/2}}{P_{\text{atm}}}\right] \tag{24}$$

$$= -\frac{1}{2P_{\text{atm}}} \underbrace{(2R\Delta P)^{1/2} T^{1/2} P_{\text{atm}}^{-1/2}}_{V}. \tag{25}$$

Thus,

$$\boxed{\frac{\partial V}{\partial P_{\text{atm}}} = -\frac{V}{2P_{\text{atm}}}}, \qquad \boxed{u_{V,P_{\text{atm}}} = \frac{V}{2}\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}}. \tag{26}$$

The negative sign describes the *direction* of the sensitivity: if P_{atm} increases at fixed differential pressure and temperature, the calculated velocity decreases. The magnitude determines its uncertainty contribution. If $u_{P_{\text{atm}}}/P_{\text{atm}}$ stays approximately constant, this contribution increases in proportion to V.

4 Combine the three contributions

Substitute Eqs. (12), (21), and (26) into Eq. (7):

$$u_V^2 = \left(\frac{V}{2\Delta P}u_{\Delta P}\right)^2 + \left(\frac{V}{2T}u_T\right)^2 + \left(-\frac{V}{2P_{\text{atm}}}u_{P_{\text{atm}}}\right)^2 \tag{27}$$

$$= \frac{V^2}{4}\left[\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2\right]. \tag{28}$$

Take the positive square root:

$$\boxed{u_V = \frac{V}{2}\sqrt{\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}}. \tag{29}$$

This is the **absolute combined standard uncertainty** in airspeed, with units of m/s.

4.1 Why absolute uncertainty can have a U-shaped curve

Use $\Delta P = \frac{1}{2}\rho V^2$ to rewrite the pressure contribution, retaining the temperature and atmospheric-pressure terms:

$$u_V^2 = \left(\frac{u_{\Delta P}}{\rho V}\right)^2 + \frac{V^2}{4}\left[\left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2\right]. \tag{30}$$

Define, for a simplified operating-range model,

$$A = \frac{u_{\Delta P}}{\rho}, \qquad B = \frac{1}{2}\sqrt{\left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}. \tag{31}$$

If A and B remain approximately constant as the operating speed changes,

$$\boxed{u_V = \sqrt{\frac{A^2}{V^2} + B^2V^2}}. \tag{32}$$

The A^2/V^2 term decreases with speed; the B^2V^2 term increases. Their competition can produce a minimum in *absolute* airspeed uncertainty. It is not guaranteed to occur within the tunnel's usable operating range.

To locate the mathematical minimum, minimize u_V^2:

$$\frac{d(u_V^2)}{dV} = -\frac{2A^2}{V^3} + 2B^2V = 0, \tag{33}$$

$$2B^2V = \frac{2A^2}{V^3}, \tag{34}$$

$$B^2V^4 = A^2, \tag{35}$$

$$\boxed{V_{\min,u_V} = \sqrt{\frac{A}{B}}} \qquad (A > 0,\ B > 0). \tag{36}$$

At this speed, the pressure-related and density-related contributions to *absolute* uncertainty are equal. If $B = 0$, this idealized model has no finite interior minimum.

5 Normalized airspeed uncertainty

Absolute standard uncertainty answers "How uncertain is the speed, in m/s?" For many requirements, we instead care how large the uncertainty is *relative to the speed being measured*. Define

$$\boxed{u_{V,\text{norm}} \equiv \frac{u_V}{V}}. \tag{37}$$

Divide Eq. (29) by V and cancel V/V:

$$\frac{u_V}{V} = \frac{V}{2V}\sqrt{\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2} \tag{38}$$

$$= \boxed{\frac{1}{2}\sqrt{\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}}. \tag{39}$$

Using Eq. (32), the same quantity is

$$\frac{u_V}{V} = \frac{1}{V}\sqrt{\frac{A^2}{V^2} + B^2V^2} \tag{40}$$

$$= \sqrt{\frac{1}{V^2}\left(\frac{A^2}{V^2} + B^2V^2\right)} \tag{41}$$

$$= \boxed{\sqrt{\frac{A^2}{V^4} + B^2}}. \tag{42}$$

For constant A and B, normalized uncertainty decreases as speed increases and approaches the constant floor B. Thus, even if *absolute* uncertainty rises at higher speed, *normalized* uncertainty need not rise.

Choosing a performance metric matters. A requirement such as $u_V \leq 0.15\,\text{m/s}$ is an *absolute* requirement. A requirement such as $u_V/V \leq 1\%$ is a *normalized standard uncertainty* requirement. These may lead to different acceptable speed ranges. Neither curve alone tells us which speed an aerodynamic experiment should use.

6 Characterize the existing measurement system first

For the actual tunnel, use the *installed* instruments and their documented uncertainty specifications, not the illustrative sensors later in this guide. For a range of trial airspeeds:

1. Calculate $\Delta P = \frac{1}{2}\rho V^2$.
2. Obtain the standard uncertainties $u_{\Delta P}$, u_T, and $u_{P_{\text{atm}}}$ from the specified uncertainty models.
3. Compute each contribution and the combined u_V using Eq. (29).
4. Compute normalized uncertainty u_V/V using Eq. (37).
5. Plot u_V and u_V/V against trial airspeed, identifying any uncertainty threshold stated in the assignment.

6. Check whether each speed is *within the pressure sensor's measurable range* and the tunnel's operating limits.

A pressure sensor with positive differential full scale ΔP_{FS} reaches its range limit when $\Delta P = \Delta P_{\text{FS}}$. Solving for the corresponding airspeed,

$$\frac{1}{2}\rho V^2 = \Delta P_{\text{FS}}, \tag{43}$$

$$V^2 = \frac{2\Delta P_{\text{FS}}}{\rho}, \tag{44}$$

$$\boxed{V_{\text{max,sensor}} = \sqrt{\frac{2\Delta P_{\text{FS}}}{\rho}}}. \tag{45}$$

This is a *range limit*, not an uncertainty limit. The tunnel or another component may impose a lower maximum. Do not interpret calculated uncertainty beyond the sensor's measurement range as an available operating capability.

7 Design backwards from a requirement

Suppose a new experiment *requires* airspeed measurements at $V_{\text{target}} = 10\,\text{m/s}$ with

$$\boxed{\frac{u_V}{V} \leq \epsilon}, \qquad \epsilon = 0.01. \tag{46}$$

The numerical requirement is illustrative; use the requirement assigned for your actual lab work.

7.1 What dynamic pressure must be measured?

Using $\rho = 1.2\,\text{kg/m}^3$ as an *illustrative* density,

$$\Delta P_{\text{target}} = \frac{1}{2}\rho V_{\text{target}}^2 \tag{47}$$

$$= \frac{1}{2}(1.2)(10)^2 \tag{48}$$

$$= \boxed{60\,\text{Pa}}. \tag{49}$$

7.2 What pressure uncertainty is allowable?

Start with the normalized uncertainty relation and impose the limiting value ϵ:

$$\epsilon = \frac{1}{2}\sqrt{\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}, \tag{50}$$

$$2\epsilon = \sqrt{\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}, \tag{51}$$

$$4\epsilon^2 = \left(\frac{u_{\Delta P}}{\Delta P}\right)^2 + \left(\frac{u_T}{T}\right)^2 + \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2, \tag{52}$$

$$\left(\frac{u_{\Delta P}}{\Delta P}\right)^2 = 4\epsilon^2 - \left(\frac{u_T}{T}\right)^2 - \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2. \tag{53}$$

Take the nonnegative square root and multiply by ΔP:

$$\boxed{u_{\Delta P,\max} = \Delta P_{\text{target}} \sqrt{4\epsilon^2 - \left(\frac{u_T}{T}\right)^2 - \left(\frac{u_{P_{\text{atm}}}}{P_{\text{atm}}}\right)^2}}. \tag{54}$$

If the expression under the radical is negative, the temperature and atmospheric-pressure terms already exceed the requirement; replacing *only* the differential-pressure sensor cannot solve the problem.

In the simplified case where the temperature and atmospheric-pressure terms are negligible,

$$\epsilon \approx \frac{1}{2}\frac{u_{\Delta P}}{\Delta P}, \tag{55}$$

$$u_{\Delta P,\max} \approx 2\epsilon \Delta P_{\text{target}} \tag{56}$$

$$= 2(0.01)(60\,\text{Pa}) \tag{57}$$

$$= \boxed{1.2\,\text{Pa}}. \tag{58}$$

This is an *approximate maximum allowable pressure standard uncertainty* for the illustrative 10-m/s requirement, not a claim about the installed instrument. Including temperature and atmospheric-pressure uncertainty reduces this allowance.

8 Sensor range versus low-speed capability

Suppose a family of *hypothetical* differential-pressure sensors each has a constant standard uncertainty equal to $k = 0.001$ (0.1%) of its pressure full scale:

$$\boxed{u_{\Delta P} = k \Delta P_{\text{FS}}}. \tag{59}$$

For these sensors, lowering full scale improves absolute pressure uncertainty, *but* decreases maximum measurable airspeed (Eq. (45)). The hypothetical data below also include an illustrative constant normalized density-related contribution $B = 0.005$ (0.5%) so that $u_V/V = \sqrt{(u_{\Delta P}/(\rho V^2))^2 + B^2}$.

Sensor	ΔP_{FS} (Pa)	$u_{\Delta P}$ (Pa)	$V_{\max,\text{sensor}}$ (m/s)	u_V/V at 10 m/s (%)
A	2500	2.500	64.5	2.14
B	500	0.500	28.9	0.65
C	125	0.125	14.4	0.51

Interpretation. Sensor A has the largest high-speed range but does not satisfy the hypothetical 1% requirement at 10 m/s. Sensors B and C satisfy that low-speed requirement, but their pressure ranges prohibit measuring some higher speeds. Sensor C is not universally "best" merely because it has the smallest uncertainty.

A complete operating envelope must satisfy *both*

$$\boxed{\frac{u_V}{V} \le \epsilon \qquad \text{and} \qquad \Delta P \le \Delta P_{\text{FS}}}, \tag{60}$$

as well as the physical tunnel and other instrument constraints. If the experiment requires both 10 m/s and 40 m/s, none of these three illustrative sensors satisfies both conditions by itself. The measurement system might require a different sensor or multiple measurement ranges.

Design takeaway. Start with the *experiment's required speeds and acceptable uncertainty.* Determine the necessary measurement capability, then choose instrumentation that meets those requirements across the needed operating range. The goal is not simply to minimize an uncertainty curve.

9 Connection to lift and drag measurements

In a subsequent aerodynamic test, the force coefficients are

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}, \qquad C_D = \frac{D}{\frac{1}{2}\rho V^2 S}. \tag{61}$$

Since $\frac{1}{2}\rho V^2 = \Delta P$ for this measurement model,

$$\boxed{C_L = \frac{L}{\Delta P S}, \qquad C_D = \frac{D}{\Delta P S}}. \tag{62}$$

Meeting an *airspeed* uncertainty requirement does not automatically meet a lift- or drag-coefficient uncertainty requirement. Force uncertainty, pressure measurement, model assumptions, and reference geometry still matter. In particular, do not treat V and ρ as independent measured inputs to a coefficient calculation if V was computed from the *same* density and differential-pressure data; that would ignore their dependence.

10 Questions to check your understanding

1. The expression $\partial V/\partial \Delta P = V/(2\Delta P)$ appears to increase with V. Show explicitly why the sensitivity actually decreases approximately as $1/V$ across operating points with constant density.
2. Explain in words why the temperature contribution to *absolute* uncertainty increases with V, while its contribution to *normalized* uncertainty remains approximately constant.
3. Can absolute uncertainty increase at high speeds while normalized uncertainty continues to decrease? Use equations from this guide to justify your answer.
4. Your experiment requires 10 m/s at 1% normalized *standard* uncertainty. What quantities must you know to establish the maximum allowable pressure-sensor standard uncertainty?
5. What do you gain and lose by replacing a large-full-scale differential-pressure sensor with a smaller-full-scale sensor? Why might two sensors be needed?
6. Why is the speed of minimum u_V not automatically the best speed for a lift-and-drag experiment?

Notes on interpreting specifications

A manufacturer's "accuracy," "precision," "resolution," and "% of full scale" are not automatically interchangeable with a *standard uncertainty.* Convert or combine specifications using the conventions supplied in your assignment, and document your assumptions. This guide uses first-order propagation and independent inputs; correlations, speed-dependent sensor behavior, offsets, calibration, and model discrepancies may require a more complete analysis. A 1% standard uncertainty is *not* the same as a 1% 95%-coverage expanded uncertainty.